

Ayush Patwa

AI/ML — Data Science — Full-Stack Developer

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Technical Skills

Languages:	Python, Java, JavaScript, SQL
Frameworks & Libraries:	React.js, Next.js, Node.js, Express.js, FastAPI, Flask
AI / Data Science:	ML, NLP, Computer Vision, Scikit-learn, Pandas, NumPy, OpenCV, Transformers
Databases:	MongoDB, Firebase
Developer Tools:	Git, GitHub, VS Code, Docker, Postman
Concepts:	Data Structures & Algorithms, REST APIs, OOP, Web Security

Projects

KisanBuddy – AI-Powered Agricultural Assistant

(Next.js, FastAPI, Gemini API, Vertex AI, Firebase)

- Developed a multilingual AI-powered Progressive Web App (PWA) to assist farmers with crop disease diagnosis, weather hazard alerts, and real-time market price intelligence.
- Implemented an agentic Planner–Executor–Validator architecture integrating Vertex AI, SWIN Transformer, LSTM forecasting, and Earth Engine services for intelligent decision making.
- Designed an offline-first architecture using IndexedDB with AES-256 encryption and background synchronization to improve reliability in low-connectivity regions.

Scantinel – Automated Web Vulnerability Scanner

(React, Node.js, Express.js, MongoDB)

- Built a full-stack web vulnerability scanner performing 10 automated security assessment phases including header analysis, SSL validation, port scanning, and exposed resource detection.
- Implemented real-time scan progress using Server-Sent Events (SSE), AI-assisted security recommendations, PDF/CSV report generation, and historical scan management.
- Developed an interactive security dashboard with risk grading, severity visualization, scan comparison, and exportable audit reports.

VeriGraph AI – Propaganda & Claim Verification Platform

(Python, FastAPI, Next.js, Transformers)

- Developed an AI-driven misinformation analysis platform capable of tracking claim propagation across News RSS, Facebook, Telegram, GDELT, and CommonCrawl sources.
- Implemented real-time propagation analytics including virality scoring, threat assessment, amplification tracking, and confidence-based claim verification.
- Designed scalable backend services with parallel data collection pipelines and interactive visualization dashboards for misinformation monitoring.

PhishGuard AI – Intelligent Phishing Detection System

(Python, Flask, Scikit-learn)

- Built an Explainable AI-powered phishing detection platform capable of identifying malicious URLs and emails using machine learning models.
- Integrated threat visualization, IP geolocation, explainable predictions (XAI), and detailed security reports to improve cybersecurity awareness.
- Designed an interactive dashboard with REST APIs for real-time phishing analysis and threat intelligence.

Achievements

- Runner-up – HackSprint 2026** — Won 2nd place and a cash prize of 3,000 for developing an AI-based solution.
- Participated in **Smart India Hackathon (SIH)**, **HackIndia**, and **HackBio**, collaborating on AI, cybersecurity, and software engineering projects.
- Created technical programming content on **YouTube**, sharing software development and technology concepts.

Certifications

- Artificial Intelligence and Machine Learning – Coding Spoon

Education

Rungta College of Engineering and Technology, Bhilai

2024 – Present

Bachelor of Technology (B.Tech.) in Computer Science

CGPA: 7.50/10